Developing an assay to titrate and differentiate metal binding sites on single atom ceria catalysts

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A novel method has been developed to estimate the extraction of the active species from a ceria-supported palladium single atom catalyst (SAC) using dimethylglyoxime (DMG). Palladium and DMG form the $Pd(DMG)_2$ complex with enough free energy release to break the Pd-O-Ce bonds anchoring the Pd ions to the catalyst. Palladium is extracted at different temperatures varying from 30ºC to 70ºC and analyzed in UV – VIS – NIR spectrometer. The initial weight loading of Pd in ceria was measured by ICP – OES..A rudimentary Van 't Hoff analysis shows a very small enthalpic contribution

Keywords *Catalysis, Dimethylglyoxime, Palladium, Chloroform, Single atom catalyst, ICP – OES, UV – Vis, Ceria*

Introduction

Since its inception in the late eighteenth century, catalysis has played a crucial role in the chemical industry. Catalytic methods are now used to manufacture oil, synthesize the majority of pharmaceuticals, chemicals, and polymers, and eliminate emissions from automobiles and power plants.[i] [ii] Around the same time, several chemists were developing new processes with heterogeneous catalysts, the most notable of which was the Haber, Bosch, and Mittasch process for the synthesis of ammonia.[iii]

Since catalysis occurs on a surface, any metal atoms that are not accessible to the reactant molecules are lost to a significant degree, and the fraction of these metal atoms should be as small as possible. The primary main focus has been to use smaller metal particles with high surface – to – volume ratios where the metal nanoparticles (NPs) are normally distributed, ideally anchored, onto high surface area and robust supports to avoid aggregation during synthesis or catalytic reaction, so that the reactant molecules can access as many surface atoms of the metal as possible.[iv] [v]

Metal NPs' catalytic behavior is modified by size reduction through cluster configurations, in which the precise arrangement of atom positions and the number of atoms inside a cluster can dramatically alter their physicochemical properties.[vi] [vii] [viii] Many attempts have been made to downsize the metal nanoclusters to isolated single metal atoms in the most effective way to use any metal atom in a supported metal catalyst, as long as the supported single metal atoms can produce desired catalytic properties. Small metal clusters can be viewed as being built up from single atoms, dimers, trimers, and so on, and they should behave more like atoms or molecules compared to surface atoms on large metal NPs, which behave like single-crystal surfaces.[ix] [x] The main differentiating parameter used for small and large NPs is dispersion, defined as:

$$\frac{\text{number of surface atoms}}{\text{total number of atoms within a particle}}$$

The eventual extrapolation of downsizing the metal nanoclusters lead to the field of single atom catalysis (SAC) which has exploded in the last 5 years. A single atom catalyst is comprised of isolated single atoms dispersed on a support. There is no spatial ordering or other significant interactions between the isolated individual atoms. Single metal atoms, as well as immediate neighboring atoms of the support surface or other functional species, make up the active sites.[xi] [xii] Depending on the relationship between a single metal atom and its neighboring atoms, the catalytic properties of individual active sites may be similar or different.[xiii]

A word of caution is to not confuse SACs with Single – Site Heterogeneous Catalyst (SSHC)[xiv] which may consist of one or more atoms in its single site. When all of the individual atoms in a SAC have similar or identical catalytic activity, the SAC can be regarded as a SSHC. SAC can also be considered as a subclass of Atomically Dispersed Supported Metal Catalyst (ADSMC) with the supported metal's dispersion[xv] considered to be 100% but, SAC's structure and catalytic performance are much better defined than an ADSMC's.

It is emphasized here that the abbreviation "SAC" only refers to supported single atoms, and that the catalytically active center of a SAC is always an isolated individual metal atom, though it may include nearby atoms from the support surface or surface functional groups. This nomenclature is restricted because it excludes the metal's bonded groups (ligands), which include the support.

Recognizing that no one term is generally accepted, we proceed with name SAC.

The support considered here., ceria/ Cerium(IV) oxide comes under the class of metal oxides, which are the most common industrial catalyst supports for bonding to atomically distributed metals, is also an active support that can provide lattice oxygen to facilitate oxidation reactions.[xvi]

The analysis of supported single metal atom catalysis is expected to shed light on the essence of heterogeneous catalysis and serve as a bridge between heterogeneous and homogeneous catalysis. Since the past decade more SACs have been developed, more catalytic reactions have been studied, and more Density – functional theory (DFT) calculations[xvii] have been performed to better understand the interactions (e.g., charge transfer) between single metal atoms and their supports, as well as the catalytic cycles on single – atom active centers which allows for higher metal loadings and improved flexibility by supplying chemically distinct neighboring atoms.

However, the stabilization of species with precise nuclearity, as well as distinguishing between them, is a major challenge. Due to the difficulty of experimentally verifying the structures, an increasing reliance on computational methods, primarily based on density functional theory (DFT), to support hypotheses for observed reactivity has developed.[xviii] But, to move forward in the design, further progress in controlled synthesis and characterization is needed.

We will specifically address the extraction and kinetics of the isolated single metal atoms distributed onto supports, such as metal oxides, in this standpoint. This Perspective is not meant to be a comprehensive overview of the area; rather, it serves as a glimpse of a rapidly expanding research field in heterogeneous catalysis. It broadcasts the catalysis community on the variety of catalytic reactions performed on SACs, new methods to anchor single metal atoms, and fundamental understanding of the catalytic mechanisms of SACs and it also aims to clear up some myths or misunderstandings about SACs and summarize the broad range of SAC applications in various catalytic reactions and provide a potential outlook on this rapidly evolving research field.

Experimental

Heterogenous catalysis research is related to the study of Transition metal ions (TMI)[xix] which includes palladium and Ceria[xx] for the application of UV – VIS – NIR spectroscopy and microscopy. Ceria can provide surface oxygen anions and OH groups as ligands, as well as lone electron pairs on oxygen for dative coordination.

Chemicals

For the synthesis of SAC-Pd/CeO_2, a palladium precursor Tetraaminepalladium(II) chloride was taken because of its high solubility in water and it does not undergo hydrolysis. This facilitates the metal ions to adsorb on the surface with a large number of nucleation centers decreasing the size of resulting NPs.[xxi]

$$\mathrm{Pd\,(NH_3)_4\,Cl_2} + \mathrm{CeO_2} \rightarrow \mathrm{Pd\,(NH_3)_4^{2+}} - \mathrm{CeO_2^{\delta-}}$$

Where $\delta -$ indicates the negative charge on the surface at high pH.

Calcining the above sample, we get $\mathrm{PdO} - \mathrm{CeO_2}$.

$$\mathrm{Pd\,(NH_3)_4^{2+}} - \mathrm{CeO_2^{\delta-}} \xrightarrow{\Delta,\ O_2} \mathrm{Pd} - \mathrm{CeO_2}$$

For the extraction of Pd from ceria, solutions of dimethylglyoxime (DMG) in ethanol (in volume), De – Ionized water (in volume), Chloroform ($CHCl_3$) (in volume) and calcined Pd in CeO_2 (in weight) were used.

Method

2 ml of DMG in ethanol and 2 ml of DI water was added to 100 mg of Pd – Ceria and sonicated for 30 min. The sonicated solution was again heated at 30ºC for another 30 min simultaneously stirring at 150 rpm.

Note here that the heating was done at temperatures ranging from 30ºC to 70ºC varying by a factor of 10ºC. We had to keep the upper limit of temperature to be 70°C because of the boiling point of ethanol (78.4ºC).

The heated solution was cooled down to room temperature before adding 2 ml of chloroform to the solution. The solution was sonicated again for 20 min. The sonicated solution is centrifuged at a high RPM rate of around 8000 – 9000 for 20 mins for sufficient separation. We observe two distinct layers after centrifugation with the bottom heavier layer containing the Pd in Chloroform. We also observe some residue at the bottom and the interface between the layers.

We extracted the bottom layer which was used to produce the calibration curve using the UV – VIS – NIR spectroscopy. We used a spectral range of 300 nm – 600 nm which is in the UV and near VIS spectrum.

Instruments

The instruments used for the above procedure:

- Glass vials
- Sonicator water bath
- Hot plate with stirrer and Oil bath
- Centrifuge tubes

— High RPM Benchtop Centrifuge
— UV – VIS – NIR Spectrophotometer
— Micro Quartz Cuvettes with Stoppers

Results and Discussion

The Pd samples analyzed in this work are divided into three categories with relevant nomenclature defined as:

- Pd_0 : Pd/CeO_2 catalyst before DMG extraction.
- Pd_e : $Pd(DMG)_2$ collected during extraction procedure into Chloroform.
- Pd_f : remaining Pd/CeO_2 after DMG extraction.

$$Pd_o/CeO_2 + DMG \overset{K}{\leftrightarrow} Pd_e(DMG)_2 + Pd_f/CeO_2$$

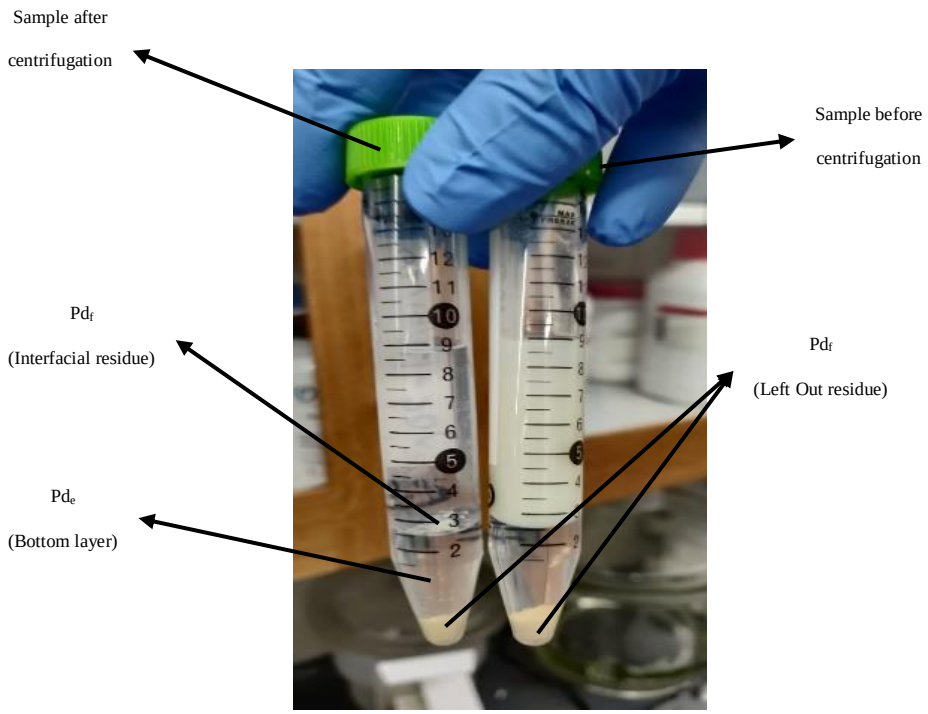
To avoid the any misinterpretation, it is stated here that the Pd_0, Pd_e and Pd_f amounts are different but still are considered as single atoms.

Following the centrifugation step, we extract the bottom layer to analyze in UV – Vis spectrometer where we generate the absorbance plots shown in the figure below.

We observe a small amount of ceria at the interface between the two layers. It is not necessarily individual crystals of ceria that are at lower density, they form an aggregate that somehow might not efficiently interact with the solvent, so they might be less mass per unit volume where we can neglect them for future calculations without any major discrepancies.

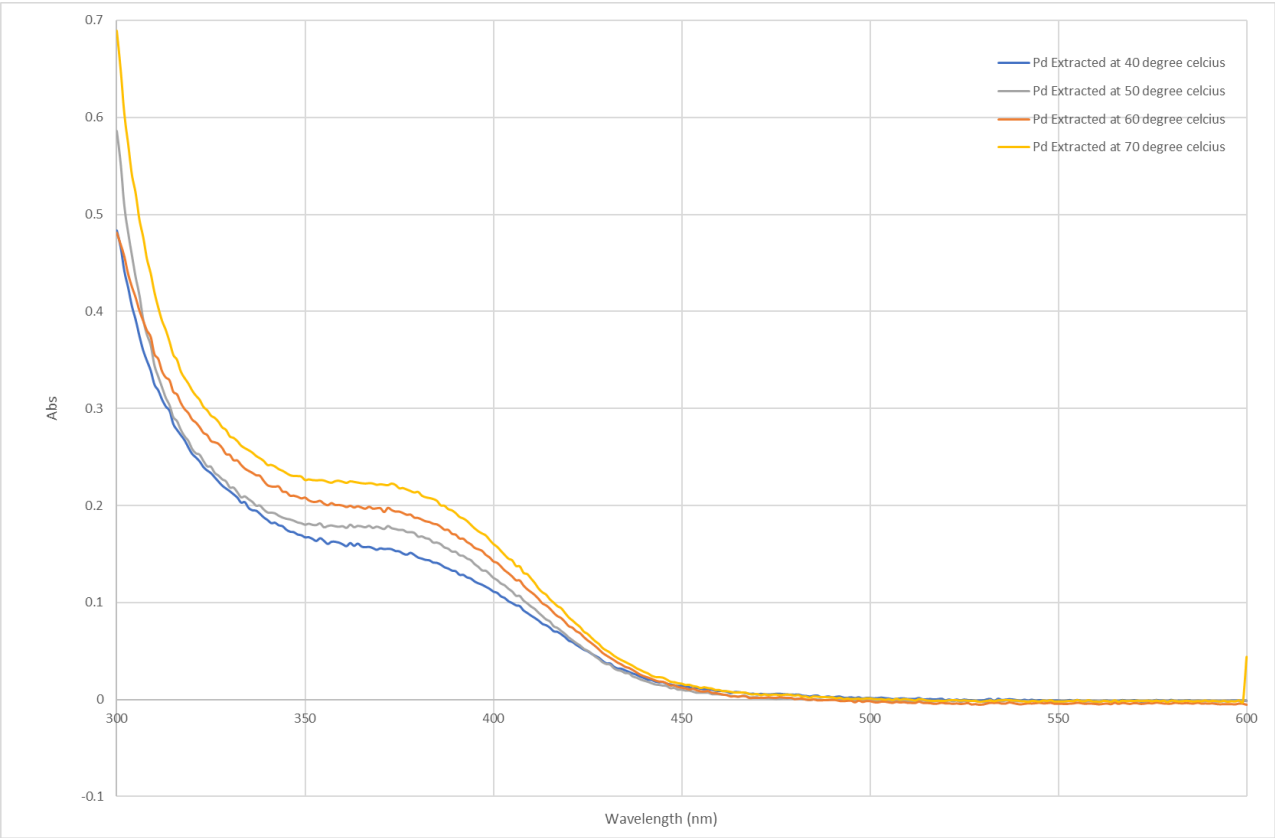
For the UV – VIS – NIR spectroscopic analyses we chose a spectral range of 265 nm to 600 nm and incorporated two types of UV plots for this paper:

— UV – Vis for varying extraction temperature
— UV – Vis for varying concentration of $Pd(DMG)_2$ in Ceria



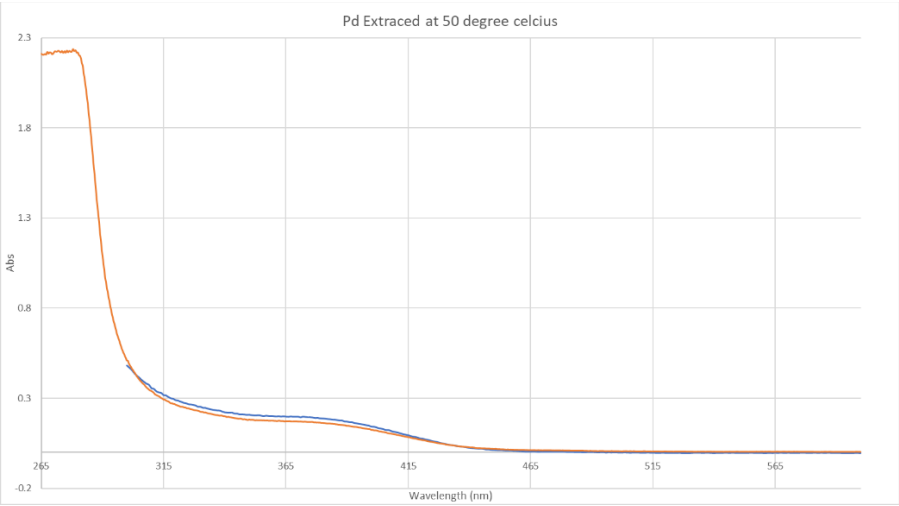
Samples of two layers with the bottom layer containing the extracted Pd into Chloroform

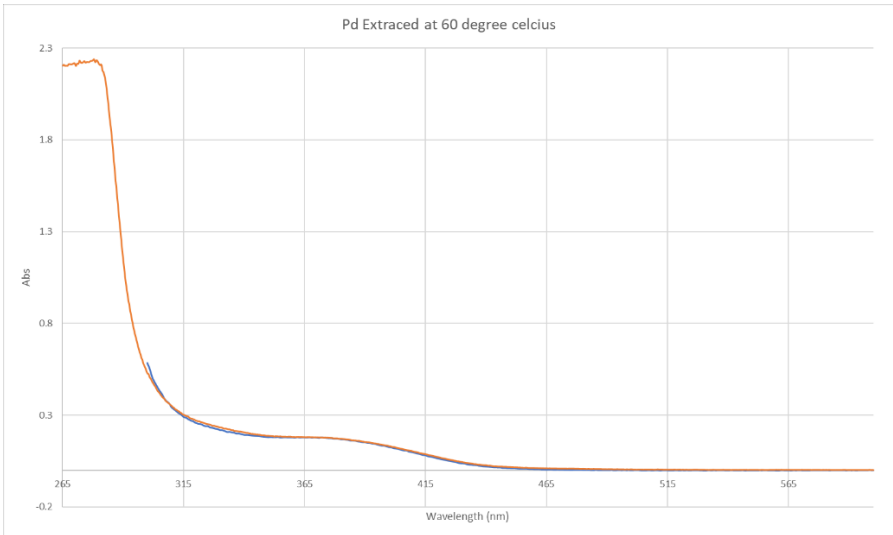
UV – Vis for Varying Temperatures:

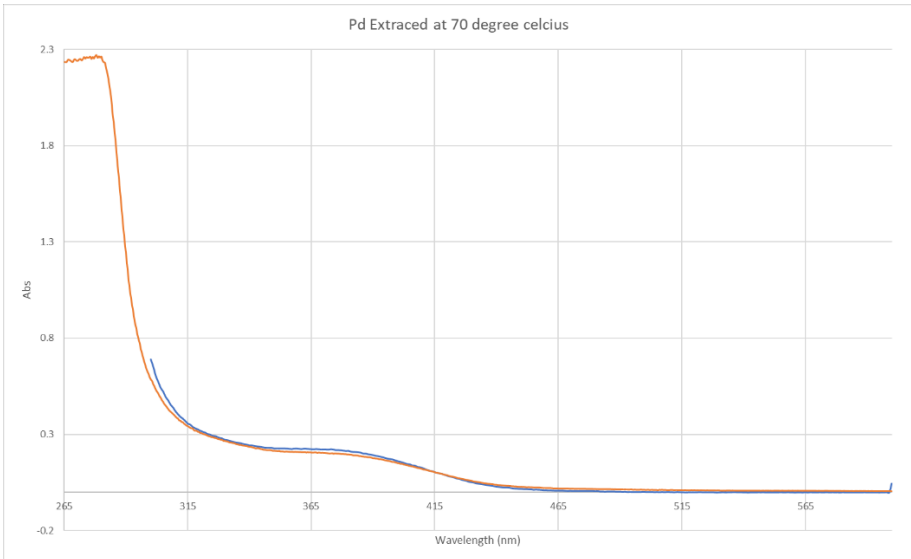


Absorbance plots of Pd at different temperatures between 300nm and 600nm

To be prudent, the samples were remade at 50ºC, 60ºC and 70ºC and run through UV – Vis between 265nm and 600nm and compared to the previous results.



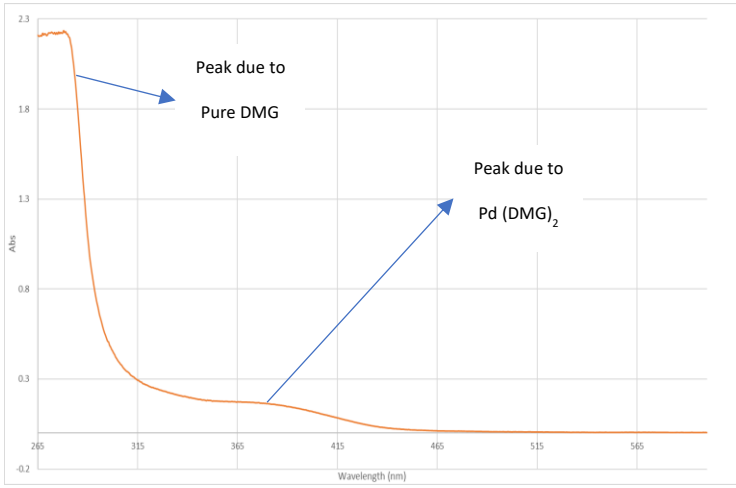




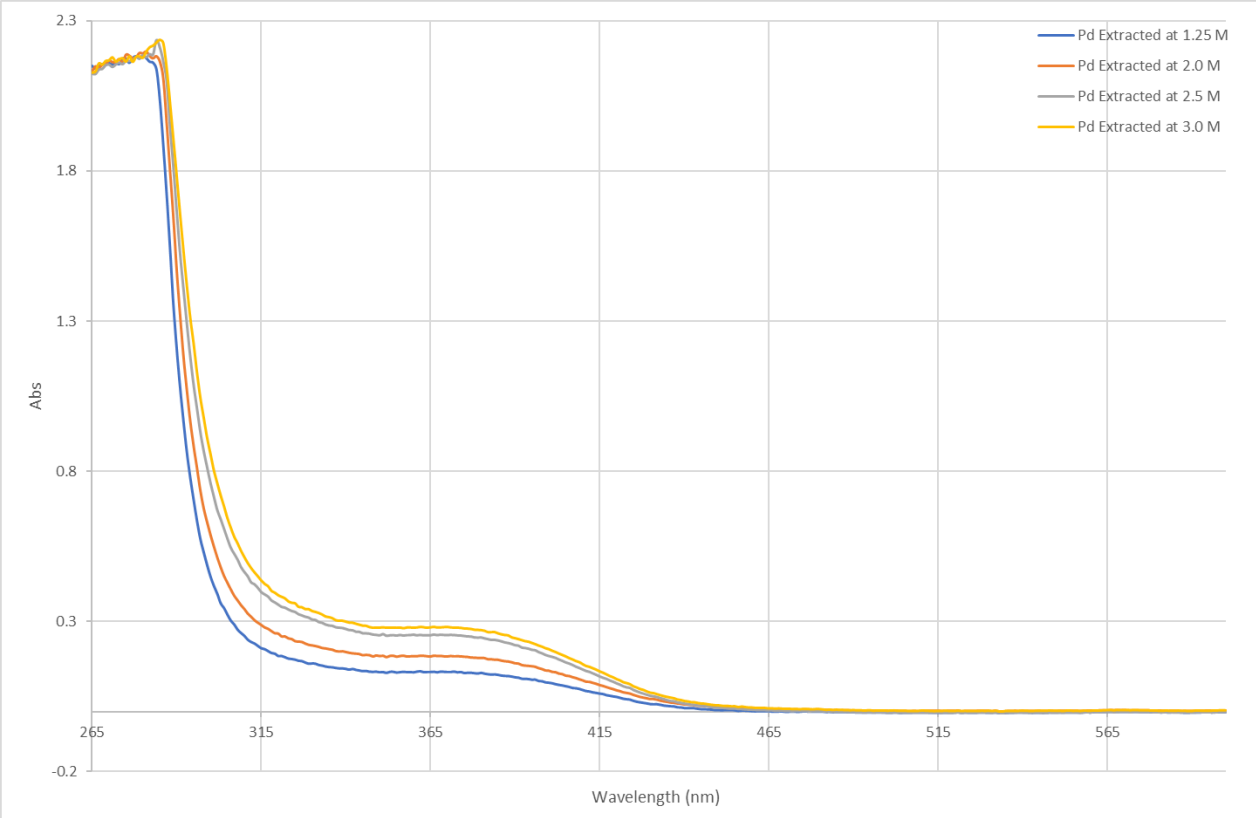
We observe that the plots are fairly close establishing the reproducibility of the data.

We get DMG in the bottom layer as, DMG is soluble in chloroform. The peak observed at ~360/370nm is the $Pd(DMG)_2$ peak. The second peak around 280/270nm is free DMG.

The peaks can be recognized by running a sample containing only DMG in chloroform; we observe a peak at the same interval of 280/270 nm. Which leads to the conclusion that the second peak is because of pure DMG while the first peak at 370/360nm is because of the $Pd(DMG)_2$ complex which only appears when the complex is formed.



UV – Vis for Varying Concentrations:



Absorbance plots of Pd at different concentrations between 265nm and 600nm

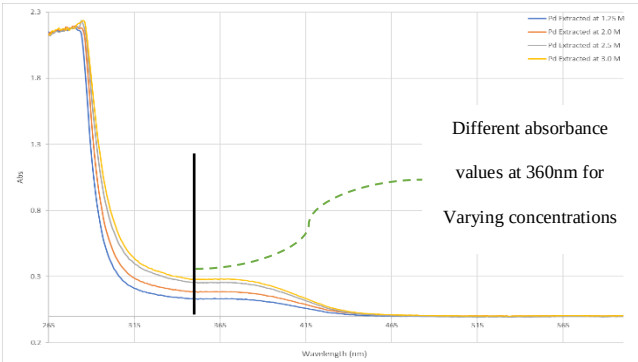
We observe $Pd(DMG)_2$ after extraction whose concentration is measured by UV-Vis spectroscopy. We determined the value of K by measuring the concentration of $Pd(DMG)_2$ at various temperatures.

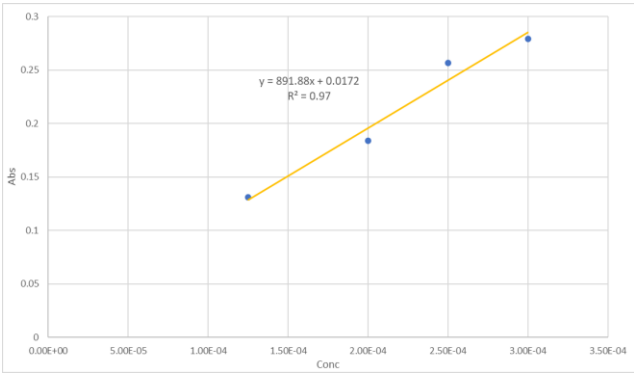
After the extraction of $Pd(DMG)_2$ we equilibrium constant of Pd extracted at different concentrations.

$$Pd/CeO_2 + DMG \overset{K_{eq}}{\leftrightarrow} Pd(DMG)_2 + Pd/CeO_2$$

$Pd - CeO_2$, CeO_2 are solids and DMG is used in large excess.

Estimating a trendline for the Absorbance vs Concentration data gives us the slope and intercept which can be used to estimate the Pd_e at different temperatures.



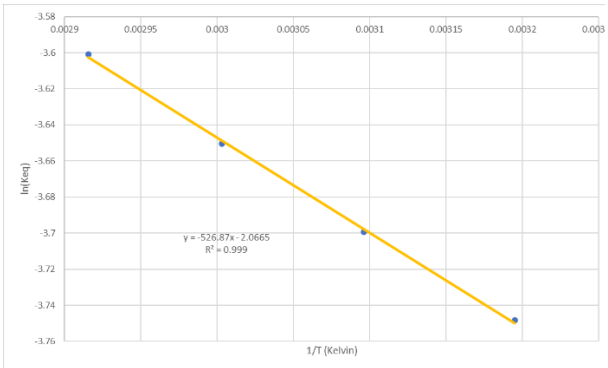


Absorbance values for different concentrations of Pd in Ceria at 360nm

Following the calculation of Pd_e we can estimate the Gibbs energy from the Van't Hoff equation:

$$\ln(K_{eq}) = -\frac{\Delta H}{RT} + \frac{\Delta S}{R}$$

$$\Delta G = -RT\ln(K_{eq})$$



Plot of ln(Keq) vs 1/T

We can estimate the enthalpy from the slope of the given plot. Calculating the Enthalpy change helped us understand the binding energy between Pd and Ceria.

$$\Delta H = 4380.396537 \text{ J/mol or } 4.38 \text{ kJ/mol}$$

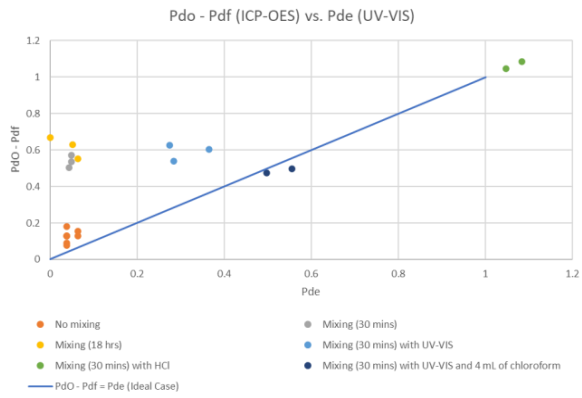
For the quantification of Pd_o and Pd_f we have run them through ICP – OES. Essentially, we should be getting:

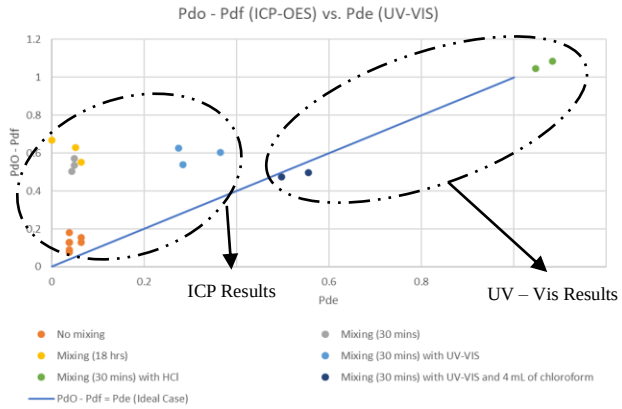
$$Pd_0 - Pd_f \cong Pd_e$$

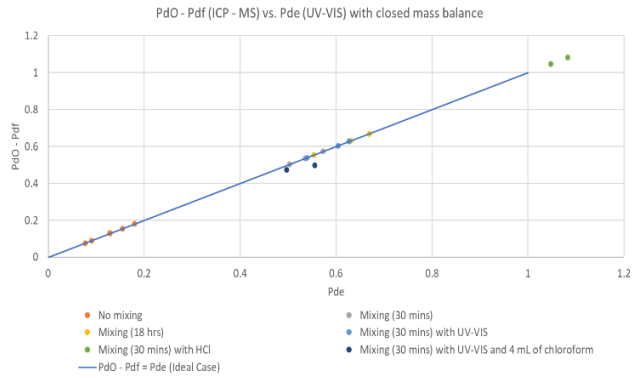
OR

$$\frac{Pd_0 - Pd_f}{Pd_e} \cong 1$$

Comparing the Pd_e results with UV – Vis and $Pd_0 - Pd_f$ with ICP gave satisfactory results. However, comparing Pd_e and $Pd_0 - Pd_f$ results with ICP gave huge discrepancies. We suspect that the lack of solubility of extracted $Pd(DMG)_2$ is responsible for the discrepancy.







The best way to help us gain knowledge of the occurrence of the peaks is from an outside source (usually DFT). But in our case, it is a little bit simpler because of our knowledge of the chemistry. All the peaks present, have to either be from the original palladium precursor $Pd(NH_3)_4Cl_2$, the $Pd(DMG)_2$ or the DMG.

In a way we are characterizing K_{eq} by measuring the concentration of $Pd(DMG)_2$ species. By measuring K_{eq} we can estimate the change in Gibbs free energy.

We have DFT predictions about how Pd binds to the surface no experimental data to confirm it. Throughout this research we directly measured the extraction and its effects on kinetics. Indirectly measured the interaction strength between Pd and CeO_2. This is something the catalysis community does not have a solid way to measure.

This begs the question whether there is only one kind of Pd (single atoms) bound to the ceria. Is there only one site or are there different sites, do those different sites have different amounts of activity? And when DMG removes the Pd, does it remove one type of site preferentially? We do not yet know the answer to this question, but now, we do have an idea of a means to tackle this problem.

Appendix

Standard Operating Procedure (UV - Vis)

i. Add 100 mg of Pd – Ceria into a glass vial.

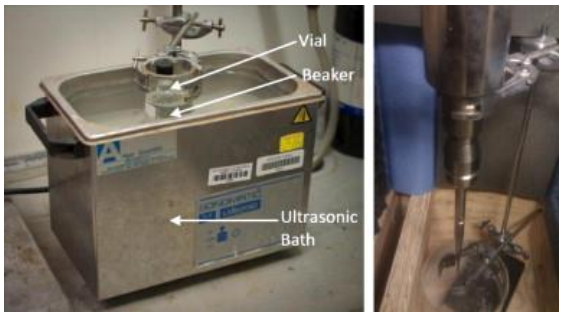
Glass vial

ii. Add 2 ml of DMG in Ethanol to the Pd – Ceria sample (should see slight orange coloring).

iii. Add 2 ml of DI water to the above solution.

iv. Sonicate the mixture in an Ultrasonic water bath for ≅ 30 min.

! Precautions *(Sonicator)*:



Sonicator water bath

The solution should be completely immersed in the water bath OR the top layer of the solution in the vial should be below the top layer of the water bath.

v. Heat the sonicated solution for 30 mins in a Hot plate heating oil bath with a stirrer at ≅ 150 RPM and 30ºC.

! Precautions *(Heating Bath)*:

Preheat the oil for around 20 mins before you put in the vial for it to reach the specified temperature.

Hot plate stirrer to adjust the temperature and RPM

Do not touch the Heating plates after the specified temperature is reached.

Oil bath for heating

The solution should be completely immersed in the oil bath OR the top layer of the solution in the vial should be below the top layer of the oil bath.

Heating plates to hold the vials

o – ring gasket for holding the vials

vi. Cool down the solution for at least 10 min after heating.

vii. Transfer the solution to a centrifuge tube.

viii. You can use more DI water to drain any more solution left in the vial to the centrifuge tube (≅ 1 – 2 ml)

ix. Add 2 ml of Chloroform into the centrifuge tube.

Centrifuge tube

x. Sonicate again for ≅ 15 min (*Note : Shake before sonicating*)
xi. Centrifuge the solution in a benchtop centrifuge at 8500 RPM for ≅ 20 min.

! Precautions *(Centrifuge)*:

Before you put in the solution; you should also put in another counterweight solution to prevent any speed wobbles due to high RPMs. The gyroscope handle most of the weight differences but any weight difference greater than few milligrams can be lethal to the operator.

High RPM Benchtop Centrifuge

The counterweight should be placed diagonally opposite to your solution. After starting the centrifuge if you feel any odd sounds, turn of immediately. Check every for any odd sounds from the centrifuge. Close the Lock lid tightly before usage.

Centrifuge rotor with Quick Lock lid and Gyroscope

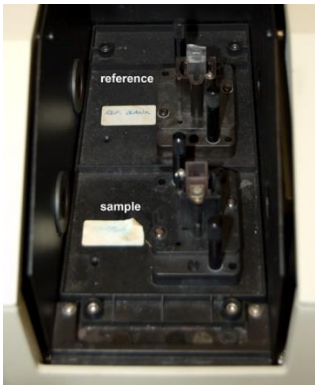
xii. After centrifugation we get two layers of liquid with the bottom layer consisting of Pd in Chloroform.
xiii. Extracted the bottom layer into a glass vial.
xiv. Repeat the above procedure for temperatures 40ºC, 50ºC, 60ºC and 70ºC. The only parameter which varies being the temperature in **step v**.
xv. We get 5 samples for different temperatures following the above procedure.
xvi. After acquiring the samples, we generate the calibration curve using UV – Vis spectroscopy (refer the standard operating procedure for UV – Vis)

! Precautions *(UV – Spectrophotometer)*:

The instrument should be on at least an hour before usage. The experiment is performed on a wavelength range or Baseline of (600 – 300 nm).

UV – VIS – NIR Spectrophotometer

You should verify that there is 0 absorbance (Abs) using no spectrophotometer cells and empty cells (if you observe an Abs > 0.005, Auto zero the value). A reference sample should be taken of only chloroform.



Sample Compartment

A Micro Quartz cuvette with a stopper should be used for the experiment to prevent Chloroform from evaporating, and it requires lesser volume of the sample. Double check the wavelength range. While using each different sample, they should be Ethanol rinsed (≅ 3 – 4 times) followed by Acetone rinse (≅ 3 – 4 times) and Air dried. Wipe the cuvettes to remove any particles before you it into the sample compartment.

Micro Quartz Cuvette with stopper 10 mm

Standard Operating Procedure (ICP–OES)

ICP preparation:

— *A 2 % (vol/vol) nitric acid in water solution needs to be created.*

- Typically, a 250 mL vol flask is used with 5 mL conc. HNO3 and 245 mL water.

— *5 standards of appropriate concentration and a blank need to be prepared according to the concentration you are expecting to see.*

- All solutions should be made day of or night before use.
- Examples would be 0.5,1,5,10, & 20 µg/mL or mg/L OR 5, 10, 50, 100, & 300 µg/mL.
- To create the standards, begin by using a reasonable concentration of standard that is not super expired and dilute.
- Make all the solutions the same volume, recommended would be 10 mL but 20 mL can be used if you will use the ICP two separate times within a day or two.
- The blank should be just 10 mL of the 2% HNO_3.
- Once all solutions are created and are 10 mL add 10-20 µL of an internal standard that will not overshadow what you are working with.
- For Silver and chloride types of yttrium has been used.
- The sample needing to be measured need to be dissolved not just suspended. So, if it is not dissolved, it should be dissolved in strong acid or base.
- Silver NP can be dissolved in concentrated nitric acid then diluted while silver chloride cannot. AgCl can only be dissolved in an ammonium solution.

! Precautions

BE VERY CAREFUL when dissolving things. It is very easy to combine things that should not be or that will explode when mixed. An easy example is ethanol and nitric acid. If working in EtOH, use the rotovap to evaporate all solvent.

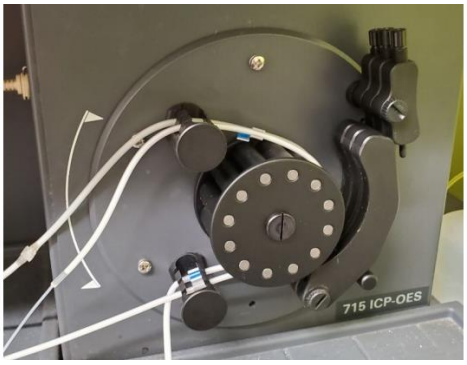
- All samples should be created and run-in duplicate. They need internal standard as well.
- At minimum 7 vials should be created at a volume of 10.01 mL or 20.02 mL.

ICP Usage:

— *15-20 minutes before you plan to use the ICP:*

- Check the liquid argon tank (outside the room) to ensure it is not empty and the flow is set to ~ 80 psi.
- Turn on the argon flow to the instrument (to the left of the instrument/ inline is on).
- Turn on the water cooler below the instrument via the large power switch at the front of the panel.
- Open the program "ICP expert II", click on 'instrument' and ensure good water and argon flow.

— *Sample collection:*



- Stretch/tighten the tubing's, one has a tiny clear flag and the other has a tiny blue flag, around the circular tube holder. You do this by using the flags to keep the tubes taught in the holder.
- Move the curved plastic pieces to the left of the wheel into place, by placing them to cradle the tubing. The pieces with the

screws can minorly adjust the tension on the tubing if flow is not correct.

- Swipe off the end of the tubing with a Kim wipe, then place it in extra 2% nitric acid.
- Turn on the plasma in the instrument tab in the computer program. A green light should come on in the window of the ICP within ~30 seconds.
- Wipe off the tube with a Kim wipe and place it into the tuning solution until after calibrations are complete.

On screen Operation:

— *Go to workbook – window – instrument setup:*

- Under status: Torch align – A Gaussian shape is desired, sharp peaks means something is wrong, misaligned/broken, or needs service. The center should be ~0 (position). This will NOT be a perfect Gaussian shape – Gaussian-ish is good enough.
- W/L (wavelength) cal – calibrate detector. This will take 2-3 minutes (you can start to set up workbook at this point).
- Calibrate wavelength, another 2-3 minutes (workbook setup during it).
- When opening a workbook, the instrument panel disappears, but it can be brought back up via window (on the top bar) – instrument setup.
- Once both the detector and wavelength is calibrated place the straw back into the 2% nitric acid.

— *Opening a workbook: New – results – click something under your name/ under a person who worked on your project – rename something else accordingly:*

- For workbooks, be sure to click new/ if you open an old result, you won't be able to change anything.

— *Under method (in Method\ Sequence\ Analysis):*

- Go to element and make sure all elements you are working with are selected (including internal standard).
- Some elements have 2 wavelengths they can work at, make sure it is the highest response wavelength.
- To the left of the elements, there is an icon showing if the element is an internal standard, analyte, or a third option. Clicking on it allows you to change what the element is analyzed as.

— *Go to conditions, the replicate read time is how long the sample is analyzed. It should be 10-20 for a normal sample, and 100 for super dilute samples. The longer you make it, the more sample is analyzed.*

- Do not touch power, any of the flow settings, or viewing height.
- Pump rate, and replicate read time and number of replicates are all you really need to worry about.
- Pump rate should be near 10 but can be adjusted.
- Replicates should be 3-5.
- Adjusting any of these settings WILL change the time your sample is analyzed. 5 replicates, 10 RPM, 10 replicate read time is ~2.5 minutes, and ~8 mL of your sample.

— *Go to standards and make sure to update the standard concentrations.*
- Always start with the most dilute and work to more concentrated.
- Standards can either be of 1 element or 2+ so long as they do not react. Ex: Cd and Se can be together but Ag and Cl can't.

— *Under Sequence make sure all samples are there and shown.*
- Highlight all the cells/samples, right click, and insert and it will copy and paste what's highlighted if necessary, to add more cells.
- On the right side, be sure to click "manual" or it will assume the autosampler is collecting at least the standards.

— *Under analysis click view (on the top bar) and select "intensity results".*
- It is better to use the intensity values and create your own calibration curve than to have the instrument approximate the concentration.

— *Next, click on 'tube' to fully highlight all of the cells then hit the play button to begin.*
- Clean the tube in the 2% HNO3 with a Kimwipe and place it into the blank as directed by the computer program.
- After 2-3 minutes (timer in bottom left) the program will prompt you for standard 1. Repeat the cleaning with a Kimwipe in-between each sample as prompted by the program until you have gone through all samples and standards.
- Often times, some samples will result in the intensity value turning red with a 'u' beside it. The 'u' stands for uncalibrated, but your values will be okay. Just plug them into the cal. Curve you make, along with any other values.
- Once you have finished, leave the tube out of any solution, if you are going to immediately turn off the plasma. If the plasma will not be turned off right away, place the straw into the 2% nitric acid again.

— *There are 2 ways to save your data:*
- Highlight and right click at the same time to copy into a notepad file and save it onto a flash drive.
- Write it by hand into your notebook.

— *To begin turning off:*
- Window – instrument – plasma off.
- Close out all programs.
- Turn off water pump and then gas line at left of instrument.
- Finally turn the gas off at the Dewar.

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References

[i] *Heck, R. M.; Farrauto, R. J. Appl. Catal., A 2001, 221, 443−45.*

[ii] *Bell, A. T. Science 2003, 299, 1688−1691.*

[iii] *S. Bordiga, A. Damin, F. Bonino, and C. Lamberti, Top. Organomet. Chem., 2005, 16, 37–68.*

[iv] *Deluga, G. A.; Salge, J. R.; Schmidt, L. D.; Verykios, X. E. Science 2004, 303, 993−997.*

[v] *Chen, M. S.; Goodman, D. W. Science 2004, 306, 252−255.*

[vi] *Zhang, J.; Sasaki, K.; Sutter, E.; Adzic, R. R. Science 2007, 315, 220−222.*

[vii] *Antolini, E. Appl. Catal., B 2009, 88, 1−24,*

[viii] *Kinoshita, K.J. Particle size effects for oxygen reduction on highly dispersed platinum in acid electrolytes. J. Electrochem. Soc. 1990, 137, 845–848.*

[ix] *Haruta, M.; Kobayashi, T.; Sano, H.; Yamada, N. Chem. Lett. 1987, 16, 405−408.*

[x] *Xu, Z.; Xiao, F.-S.; Purnell, S. K.; Alexeev, O.; Kawi, S.; Deutsch, S. E.; Gates, B. C. Nature 1994, 372, 346−348.*

[xi] *Qiao, B.; Wang, A.; Yang, X.; Allard, L. F.; Jiang, Z.; Cui, Y.; Liu, J.; Li, J.; Zhang, T. Nat. Chem. 2011, 3, 634−641.*

[xii] *Yang, X.-F.; Wang, A.; Qiao, B.; Li, J.; Liu, J.; Zhang, T. Acc. Chem. Res. 2013, 46, 1740−1748.*

[xiii] *Flytzani-Stephanopoulos, M. Acc. Chem. Res. 2014, 47, 783−792.*

[xiv] *Thomas, J. M.; Raja, R.; Lewis, D. W. Angew. Chem., Int. Ed. 2005, 44, 6456−6482.*

[xv] *Nie, L.; Mei, D.; Xiong, H.; Peng, B.; Ren, Z.; Hernandez, X.I.P.; DeLaRiva, A.; Wang, M.; Engelhard, M.H.; Kovarik, L.; et al. Activation of surface lattice oxygen in single-atom Pt/CeO2 for low-temperature CO oxidation. Science 2017, 358, 1419–1423.*

[xvi] *Safonova, O.V.; Guda, A.A.; Paun, C.; Smolentsev, N.; Abdala, P.M.; Smolentsev, G.; Nachtegaal, M.; Szlachetko, J.; Soldatov, M.A.; Soldatov, A.V.; et al. Electronic and geometric structure of Ce3+ forming under reducing conditions in shaped ceria nanoparticles promoted by platinum. J. Phys. Chem. C 2014, 118, 1974–1982.*

[xvii] *Spezzati, G.; Su, Y.; Hofmann, J.P.; Benavidez, A.D.; DeLaRiva, A.T.; McCabe, J.; Datye, A.K.; Hensen, E.J.M. Atomically dispersed Pd–O species on CeO_2 (111) as highly active sites for low-temperature CO oxidation. ACS Catal. 2017, 7, 6887–6891.*

[xviii] *Vorobyeva, E. et al. Atom-by-atom resolution of structure–function relations over low-nuclearity metal catalysts. Angew. Chem. Int. Ed. 58, 8724–8729 (2019).*

[xix] *A. B. P. Lever, Inorganic Electronic Spectroscopy, Elsevier, Amsterdam, 2nd edn, 1984.*

[xx] *A. A. Kaplyanski, Spectroscopy of Solids containing Rare Earth Ions, Elsevier, Amsterdam, 1987.*

[xxi] *Xu, J.; Li, L.; Li, G. A facile approach to well-dispersible CeO_2 nanoparticles. J. Dispers. Sci. Technol. 2008, 29, 1072–1076.*